Calling a Claude model through Amazon Bedrock

INSTRUCTOR ANSWERS - STAFF ONLY

Core questions 1-5

Use groups of five at suitable points in the session. Ask learners to explain their choice and one rejected option. Use the final five if helpful. Keep these group discussions separate from the existing individual knowledge-check scores.

1. Kestrel asks whether the assistant could read photographs of router labels that customers send. How does this requirement affect the choice of model?

Correct answer: A. It rules out every candidate without image input before any evaluation of answer quality.

Modality is the first requirement in the identify step: a model that can't accept the input that the work needs is ruled out at once. Testing on Kestrel's cases then compares only the candidates that meet every requirement. The course model accepts image input, so it would stay on the list.

2. Two candidate models score similarly on a published benchmark. Why does the course still evaluate them on Kestrel's own cases?

Correct answer: A. Benchmarks can use other tasks and data, so only Kestrel's cases show quality, latency and cost.

The evaluate step tests whether each candidate can do Kestrel's work, such as summarising its cases correctly, with acceptable latency and cost. Published benchmarks may use other tasks and data, so they can't settle that question. The course uses prompts and retrieval, not fine-tuning, so no candidate is trained before the comparison.

3. Kestrel wants summaries of each day's closed cases by the next morning, at the lowest price per call. Which way of calling the model suits this work?

Correct answer: B. Batch inference suits it, because it reads requests from S3 and writes the results to S3 later.

Batch inference processes requests asynchronously from files in S3, which suits work that no one waits for, and the course's pricing file records a batch price of half the on-demand price for the course model. The team still checks that batch inference supports the model and Region. Streaming changes when text arrives, not the price per token, and a short shared prefix is too small to cache.

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4. A team setting up a new AWS account finds that the model it planned to use is in the Legacy state. What does this mean for the team?

Correct answer: A. The team can't adopt the model as a new customer, so it should choose an Active model instead.

A Legacy model is scheduled for retirement, and new customers can't adopt it. Existing users can keep using it until its end-of-life date, although they may lose access after 15 days of inactivity; that right belongs to existing customers, not to a new account. Before setting up a new account for the course, check the model's state on its model card.

5. Kestrel's finance team wants to attribute the cost of the course model to each application. Why must an application inference profile for that model be created from the EU profile, not from the base model?

Correct answer: B. Bedrock serves the course model only through inference profiles, so the copy must be of a system profile.

An application inference profile copies a model or a system-defined profile so that you can tag its usage and attribute its cost. The course model doesn't accept on-demand calls by its base identifier, so a profile that copies it must copy a system-defined profile, such as the EU profile. Tags belong to the application profile that you create, not to the EU profile.

Discussion: Ask the group to connect one of these ideas to today's Kestrel example. Use the Day 2 learner handbook to resolve uncertainty.

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Core questions 6-10

6. A learner's call from London returns the ValidationException that asks for an inference profile. Which response to the error does the course recommend?

Correct answer: D. Check that the client sends config.MODEL_ID, and keep eu-west-2 as the source Region.

The ValidationException identifies an unsupported modelId; it isn't an IAM error, so a policy change doesn't fix it. The client should send the EU profile from config.MODEL_ID. Changing the Region could make the call work, but it can also change where Kestrel's data is processed, which is why the course keeps eu-west-2.

7. AWS opens a new European Region that can serve the course model. How does this affect calls that the assistant sends through the existing EU profile?

Correct answer: C. Its destinations stay the same, and the team would change identifier to use a new profile.

The destination list of a geographic profile never changes. AWS may create a new profile that includes the new Region, and the team would change its identifier to use that profile. A global profile's destination list, by contrast, can grow as AWS adds commercial Regions, which is why the claim that the EU profile adds Regions automatically is wrong.

8. The global profile for the course model costs about 10 per cent less than the EU profile. Why does the course use the EU profile for Kestrel?

Correct answer: A. Kestrel's data must stay in Europe, and the global profile could process calls in any commercial Region.

A geographic profile keeps processing inside its geography, and a global profile can process a call in any supported commercial Region. Kestrel's rule that its data stays in Europe therefore decides the choice, and the course accepts the higher price. London lists both profiles for the course model, and both charge the price of the Region that you call.

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9. A colleague proposes changing MODEL_ID in config.py to a different profile. What does MODEL_CHOICE_REASON in the same file help the colleague do?

Correct answer: B. It records the requirement that the choice meets, so a change can be checked against it.

The comment and MODEL_CHOICE_REASON explain that the EU profile keeps Kestrel's data in Europe, a requirement that any future change must preserve. Keeping the identifier in one place also makes a model change one edit that the same tests can check. Bedrock and IAM never read the file, and a sample that omits the identifier uses the framework's own default model.

10. A model appears in aws bedrock list-foundation-models for London, but calls to it fail with AccessDeniedException. What should the team check next?

Correct answer: D. Check the account's agreement status, because a catalogue entry shows only what the Region offers.

A catalogue entry shows that Bedrock offers the model in that Region; the account also needs the provider agreement before it can call the model. aws bedrock get-foundation-model-availability reports agreementAvailability.status, which Day 1 taught. An exceeded quota returns ThrottlingException, not an access error.

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Core questions 11-15

11. A learner sets both temperature and topP in one Converse request to Claude Haiku 4.5. What does the handbook say about this?

Correct answer: C. The model accepts one setting or the other, not both, so the course client sets only temperature.

Claude Haiku 4.5 accepts temperature or topP, but not both in the same request, so the course client sets only temperature, at 0. Both are base inference settings in inferenceConfig. A parameter that Converse doesn't define, such as Claude's top_k, is the kind that goes in additionalModelRequestFields.

12. The course client sets MAX_ATTEMPTS to 6, BACKOFF_BASE_SECONDS to 2 and BACKOFF_CAP_SECONDS to 32. What is the longest that one call to converse() can spend waiting between attempts before it gives up?

Correct answer: D. It can wait up to 62 seconds, the sum of the five ceilings from 2 to 32 seconds.

The client waits only between attempts, so six attempts give five waits, with ceilings of 2, 4, 8, 16 and 32 seconds. Full jitter waits a random fraction of each ceiling, so 62 seconds is the most, and the usual total is less. The cap limits each wait, not the total, and there is no wait after the last attempt.

13. A developer suggests calling InvokeModel with Claude's own request body instead of Converse. What would Kestrel lose?

Correct answer: C. It would lose portability, because code for one provider's body doesn't move to another model.

InvokeModel takes a request body in the provider's own format, with fields such as anthropic_version, so the code is tied to that provider. Converse uses one request and response shape for every model that supports messages, which is why the course uses it. InvokeModelWithResponseStream exists, and both operations need the same IAM actions as Converse.

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14. The Converse API defines nine stop reasons. Which three does the course client treat as a complete reply?

Correct answer: D. It treats end_turn, stop_sequence and tool_use as complete, and records every other one as rejected.

end_turn means that the model finished, stop_sequence means that it produced one of your stop sequences, and tool_use means that it asks the application to run a tool. A reply that stops at max_tokens can look complete but has been cut off, so the client rejects it. Guardrail and content-filter stops change or block the content, so they are rejected too.

15. Why does the course client select content blocks by type instead of reading message["content"][0]["text"] directly?

Correct answer: D. The first block isn't always a text block, and a reply that requests a tool holds a toolUse block.

A model's reply can hold several blocks of different types. On Day 3, a reply that asks for a tool holds a toolUse block, which has no text field, so reading the first block directly would fail. The result's text() method joins the text blocks and raises an error when there are none.

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Core questions 16-20

16. A call returns ModelErrorException with HTTP status 424. What does the course client do?

Correct answer: B. It stops and records the attempt, because the code isn't one of the four in RETRYABLE_CODES.

The client retries only ThrottlingException, ServiceUnavailableException, InternalServerException and ModelNotReadyException. ModelErrorException means that the model failed while it processed the request, and the client stops and records the attempt. Returning an error message as if it were an answer would hide the failure from the caller.

17. A call with 2,000 input tokens and 300 output tokens uses the course model at 1.10 USD per million input tokens and 5.50 USD per million output tokens. What does the call cost?

Correct answer: A. It costs 0.00385 USD, from 2,000 tokens at the input price and 300 at the output price.

Bedrock prices input and output tokens separately: 2,000 × 1.10 ÷ 1,000,000 is 0.0022 USD, and 300 × 5.50 ÷ 1,000,000 is 0.00165 USD, so the call costs 0.00385 USD. Pricing every token at the input rate ignores the output price, which is five times higher. The ledger compares its cost_usd with this calculation.

18. The client stops waiting for a response after its read timeout, so the attempt has no usage. How does the ledger record the attempt's cost?

Correct answer: C. It records a null cost and sets usage_known to false, because tokens may have been generated.

An attempt with no usage gets a null cost and the cost status unknown_usage, because the model may have generated tokens that Kestrel will pay for. A cost of zero would understate the spending and hide the gap from a reviewer. Every attempt, including a failed or timed-out one, has its own ledger row.

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19. A learner adds a cache checkpoint after Kestrel's system prompt and its two tool definitions, but the usage shows no cache fields. What is the most likely explanation?

Correct answer: C. The prefix is shorter than the 4,096-token minimum for the course model, so Bedrock doesn't cache it.

For explicit caching, the prefix before a checkpoint must reach a minimum size, which was 4,096 tokens for Claude Haiku 4.5. A checkpoint after a shorter prefix doesn't fail the call, but nothing is cached. Claude models support only explicit caching, and caching works with on-demand calls, not with batch inference.

20. A summary of case_1028 returns with the stop reason end_turn and passes CaseSummary validation, but its case_id is case_1003. Why does the application also compare the identifier with the case it asked about?

Correct answer: D. A schema checks only the answer's shape, so a valid summary can still describe the wrong case.

The schema accepts case_1003 because it has the right pattern and shape, but the summary would name the wrong case. Validation proves the form of an answer, not that it answers the question asked, so the application adds a check on the identifier. end_turn means that the model finished its reply normally.

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Optional extra questions 21-25

21. A learner wants Bedrock's native structured output for CaseSummary. Which part of a schema would Bedrock refuse?

Correct answer: B. It would refuse maxLength on next_step, because native output doesn't support length limits.

Native structured output refuses string length limits such as minLength and maxLength, and numeric limits such as minimum and maximum, which is why CaseSummary checks lengths in validators. Every object must set additionalProperties to false, or Bedrock refuses the request. Bedrock supports pattern, so checking the identifier in a validator is the course's choice.

22. Code, not a person, consumes the assistant's case summaries. Why does the course advise against streaming those answers?

Correct answer: D. Code needs the whole answer and its stop reason before it validates, and they arrive at the end.

In a stream, the stop reason and the usage arrive only in the final messageStop and metadata events. Code can't validate an answer until it has the whole text and knows why the model stopped, so streaming adds complexity without a benefit. The example learner policy allows both invocation actions, and a stream is priced by the same tokens.

23. The demonstration injects a ThrottlingException with botocore's Stubber, then a stubbed reply, and the client recovers. What does this result prove?

Correct answer: A. It proves that the client's own error handling, retry decision and ledger rows work as designed.

Stubber returns queued responses without contacting Bedrock, so the test exercises only the client's deterministic code. An injected failure is never evidence of how Bedrock or the model behaves, and the stubbed reply has no tokens. The injected rows carry run identifiers that start with injected-, which separate them from live attempts.

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24. Why does the course client turn off boto3's own retries with total_max_attempts set to 1?

Correct answer: B. SDK retries happen inside client.converse(), so the ledger couldn't record each attempt separately.

When boto3 retries inside client.converse(), the application receives one result and can't write a ledger row for each internal attempt. The course turns the SDK's retries off and runs its own loop, which records every attempt. boto3's standard mode does use exponential backoff, so that distractor misdescribes it.

25. A request for a long answer raises ReadTimeoutError. Why doesn't the client retry it?

Correct answer: C. The client stopped waiting, but the request may have run, so a retry could incur a second charge.

A read timeout means that the client stopped waiting, not that Bedrock stopped working, so the first request may have been processed and charged. The client records the attempt as timed_out with unknown usage and leaves the next decision to its caller. The course turns off SDK retries, so no hidden retries happened inside the call.

Discussion: Ask the group to connect one of these ideas to today's Kestrel example. Use the Day 2 learner handbook to resolve uncertainty.